IS113 – WEEK 10 ACTIVITY

DIGITAL INNOVATION STRATEGY ANALYSIS

From Technology Evaluation to Innovation Strategy

Project: FLOWBIKES: A Web-Based Bicycle Sales System

Team MAAFF | Mapúa Malayan Colleges Mindanao | Selected Technology: AI / Generative AI Concierge

Project Context

FlowBikes is a bicycle sales business whose existing work has relied on manual spreadsheet logs and unorganized direct-message channels. The project planning documents identify concerns such as delayed responses to technical inquiries, risks of inventory double-booking, and outdated inventory displays. The proposed web platform centralizes bicycle listings and inventory-related records, supports AI-assisted technical inquiries and product recommendations, and routes interested buyers to direct seller communication. The approved scope includes administrator authentication only. Customer accounts, online payment, automated notifications, courier integrations, native mobile apps, and customer repair-status tracking are excluded.

PART A – INNOVATION STRATEGY

Innovation Goal

Modernize FlowBikes' customer inquiry and inventory workflows through a centralized web platform with an AI Concierge. The system is intended to make product information easier to access, support bicycle discovery, improve inventory information consistency, and reduce reliance on scattered spreadsheets and messages.

Problem or Opportunity

Manual inventory records and unorganized customer messages can make it difficult to keep product information current, coordinate inquiries, and avoid stock conflicts. A centralized catalog and AI-assisted first-line responses create an opportunity to improve information access while preserving direct communication with the shop for buyer follow-up.

Proposed Technology and Strategic Approach

The proposed technology is a generative-AI Concierge integrated with the FlowBikes web platform and relevant product/inventory information. The strategic approach is to build the core catalog and inventory workflow first, integrate AI-assisted inquiry handling and product suggestions, test the integrations, and obtain owner/user feedback before final acceptance.

- Keep AI as an assistant for product discovery and common technical questions, not as an autonomous decision-maker.
- Ground responses in approved product and inventory information where technically feasible, and make direct seller contact available.
- Apply human oversight, accuracy checks, privacy safeguards, and documented risk monitoring.
- Keep implementation within the academic timeframe and available resources, monitoring third-party API limits.

Short-Term and Long-Term Objectives

PART B – DIGITAL BUSINESS TRANSFORMATION

PART C – INNOVATION FRAMEWORK

Framework Selected: Design Thinking

Design Thinking is a user-centered framework suitable for a problem involving customer inquiry and product discovery. It encourages the team to understand user needs, define the problem, explore solutions, prototype, and test. The stages below are an application guide, not a claim that every stage has already been completed.

Why the Framework Fits

Design Thinking fits because the problem concerns the experience of customers and shop staff, not just the choice of technology. It helps connect AI to a real inquiry-handling need, test assumptions before acceptance, and refine the solution based on feedback. It also supports delivering and validating core functions before adding features without evidence of need.

PART D – BUSINESS VALUE AND COMPETITIVE ADVANTAGE

These are expected benefits, not measured results. Do not report percentage improvements or cost savings until baseline data and post-implementation evidence exist. KPIs must match features actually implemented.

AI INTEGRATION – RECOMMENDATION VALIDATION

AI-Use Disclosure

ChatGPT was used to organize the innovation strategy, map the proposed transformation, apply a user-centered framework, suggest KPIs, and draft AI validation entries. The group should verify recommendations against the approved FlowBikes scope and original project artifacts. Scholarly/framework sources should be checked through official publications before submission. AI-generated suggestions are draft support, not proof of actual results.

References

National Institute of Standards and Technology. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0). https://doi.org/10.6028/NIST.AI.100-1

UNESCO. (2021). Recommendation on the Ethics of Artificial Intelligence. https://unesdoc.unesco.org/ark:/48223/pf0000381137

FlowBikes Team MAAFF. (2026). FlowBikes Project Management Planning Portfolio. Internal project artifact.

FlowBikes Team MAAFF. (2026). Week 8 Responsible Technology Adoption: Literature Review Matrix and Conceptual Framework. Internal project artifact.

Final Review Checklist

- ■ Part A: Innovation Strategy completed.
- ■ Part B: Digital Business Transformation completed.
- ■ Part C: Innovation Framework applied and justified.
- ■ Part D: Business Value, Competitive Advantage, and KPIs included.
- ■ AI recommendations evaluated and AI use disclosed.
- ■ Project claims checked against approved FlowBikes documents.
- ■ References independently checked and document converted to PDF.

END OF ACTIVITY

Time Horizon	Objectives
Short-term	Complete approved core catalog and inventory functions; integrate the AI Concierge; test typical and edge-case inquiries; verify product suggestions and inventory claims; document defects and owner feedback.
Long-term	Maintain a centralized source of product information; improve consistency and accessibility of customer support; monitor AI response quality and service reliability; review operational indicators; consider future enhancements only when supported by evidence and owner priorities.

Area	FlowBikes Analysis
Current Process	Product and inventory information is managed through manual spreadsheet logs, while customer inquiries arrive through direct messages that may be scattered or difficult to organize. This creates risks of outdated listings, delayed replies, and inventory coordination errors.
Proposed Transformation	A centralized web catalog presents bicycle listings and specifications, supports filtering and inventory synchronization, and includes an AI Concierge for initial technical questions and product suggestions. Interested customers are routed to direct seller communication. Staff continue to handle final sales discussions and decisions.
People and Roles	Customers use the public catalog and AI assistance. Shop staff/admin maintain product and inventory records and handle buyer follow-up. The project team designs, develops, tests, and maintains the system. Staff may need orientation on accurate inventory updates, AI output review, inquiry escalation, and defect reporting.
Resources	Development computers, internet, web hosting, a database/backend service such as Supabase, AI/API access, map/weather services, source-code management, testing time, and staff participation. Use free or low-cost services where suitable and monitor API usage and availability.
Implementation Challenges	Potential barriers include inaccurate product data, incorrect AI responses, third-party API outages or limits, privacy/security requirements, integration defects, limited team availability, and academic deadlines. Mitigations include staged implementation, realistic testing, human oversight, access controls, privacy-aware data handling, and direct seller contact as a fallback.

Stage	Application to FlowBikes	Evidence / Output to Prepare
Empathize	Understand how customers search for bicycle specifications and ask technical questions, and how staff maintain inventory and respond to inquiries.	Owner/staff feedback, customer journey notes, or documented requirements. If no user research occurred, record it as proposed work.
Define	Frame the problem as fragmented product/inventory information and inconsistent handling of customer inquiries.	Approved problem statement, requirements, and scope statement.
Ideate	Consider a centralized catalog, filtering, synchronized inventory, AI product suggestions, and direct Contact Seller routing.	Feature list, alternatives considered, and reasons for selected scope.
Prototype	Build wireframes and an initial web implementation of the catalog, admin functions, and AI Concierge.	Wireframes, prototype, or working-system demonstration, as available.
Test	Check catalog accuracy, inventory updates, AI responses, usability, integrations, and escalation to seller communication.	Functional/integration tests, defect log, user feedback, and UAT evidence.

Value Area	Expected Value for FlowBikes	Possible KPI / Evidence
Operational Efficiency / Productivity	Centralized listings and inventory records may reduce repeated manual checking and make product information easier to maintain.	Percentage of active listings with complete specifications; inventory update accuracy; time required to update a listing.
Cost Savings / Resource Optimization	Hosted services and AI-assisted routine responses may reduce avoidable manual effort. Monitoring usage can help manage API costs.	Monthly hosting/API usage and cost; routine inquiries handled by AI; staff time spent on routine inquiries, if measured.
Service Quality / User Experience	Customers can browse/filter bicycles and receive initial help with product questions, with direct seller contact for follow-up.	Catalog task completion; user satisfaction; AI answer accuracy on a reviewed test set; unresolved inquiry rate.
Decision-Making	Centralized product and inventory data can help staff check stock information and respond consistently.	Inventory discrepancies; percentage of listings synchronized with database records; data corrections.
Competitive Advantage / Differentiation	A convenient catalog with AI-assisted bicycle guidance can distinguish FlowBikes from manual inquiry-only workflows.	Catalog engagement; qualified inquiries routed to seller; customer feedback; inquiry-to-contact rate, if measured.
Measurable Success Indicators	Set baselines and targets with the owner and measure after implementation rather than assuming benefits.	For each KPI, record its baseline, target, measurement period, calculation, and data source.

AI-Generated Recommendation	Validation Source	Evaluation and Revision
Use AI to assist with customer technical questions and bicycle product recommendations.	FlowBikes scope and NIST AI Risk Management Framework (2023); UNESCO Recommendation on the Ethics of Artificial Intelligence (2021).	Accepted with safeguards. Test relevance and accuracy; do not allow invented product specifications or stock claims; provide direct seller contact; apply human oversight and ongoing monitoring.
Centralize product/inventory information and monitor system/AI performance using defined indicators.	FlowBikes Project Charter, WBS, Scope Statement, Week 8 Responsible Technology Adoption review, and NIST AI RMF (2023).	Accepted. This aligns with the planned scope. Measure KPIs from actual system and operational records; do not claim benefits before data are collected.